

New antenna design

Ready during fall 2026

Revision PA1

Lannik Kanin Antenna

IFX radar

Concept

Size

- 100x60 mm
- Based on Golden Devices Antenna

Front-end

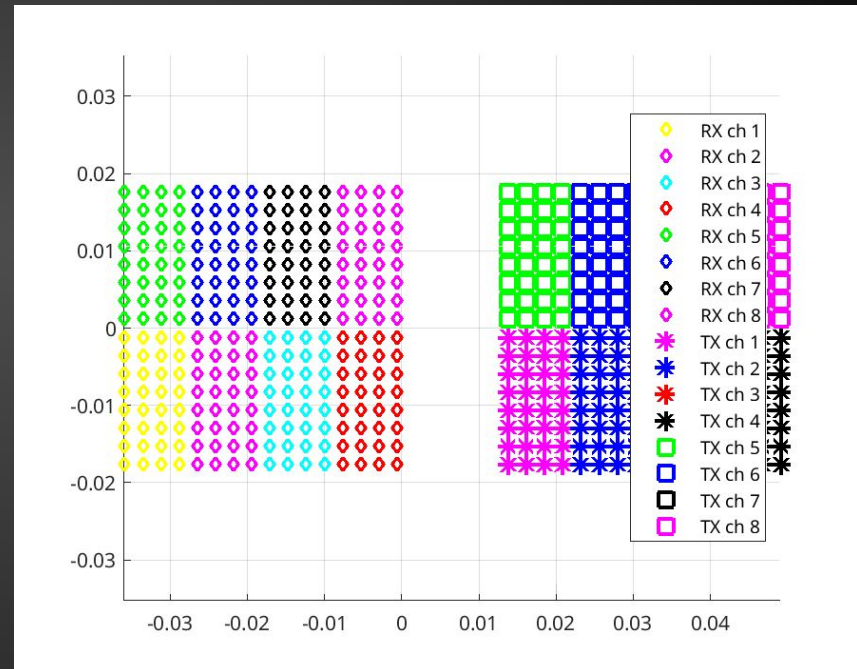
- Antenna layout =>
- 8 RX and 8 TX
 - 8 TX coherently
- Covers 1 GHz system bandwidth
- Center frequency 76.5 GHz

Chipset

- CTRX from IFX
I2C comm+ Ethernet

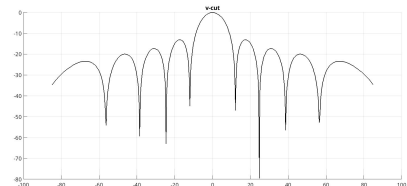
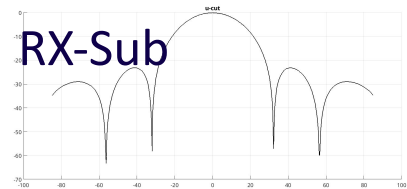
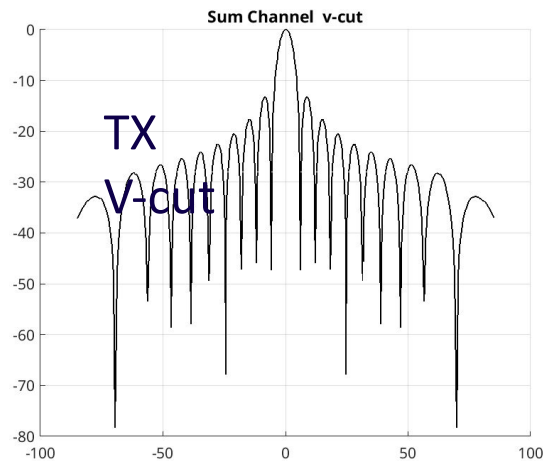
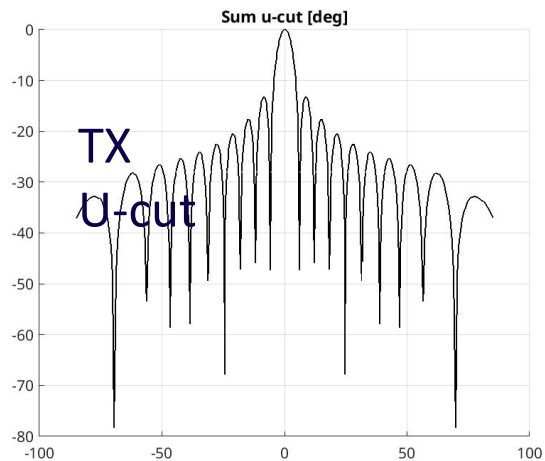
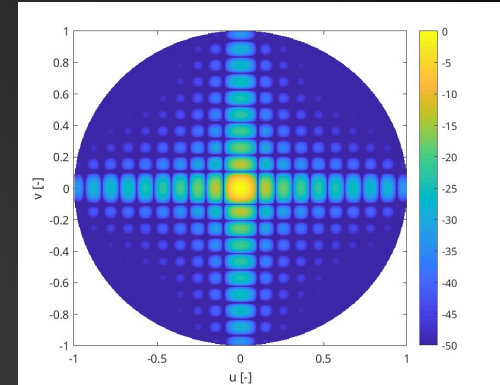


2016-09-07



Antenna Pattern Characteristics

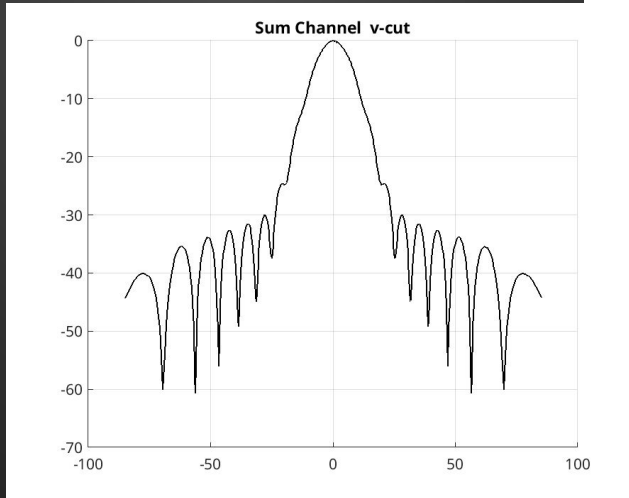
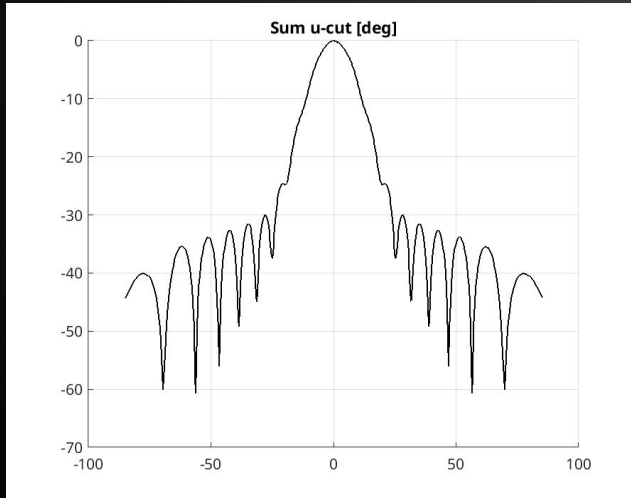
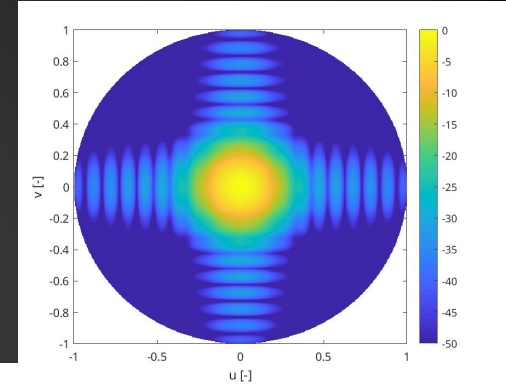
- RX & TX Beam Characteristics
 - Without phase & amplitude taper:
 - Directivity sum beam TX: ~31dBi
 - TX sum Beam (one way, 3dB):
 - Az and El: ~ ± 3 deg
- Directivity sub channel TX & RX: ~21 dBi



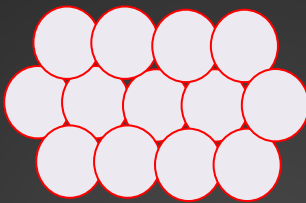


Antenna Pattern Characteristics (modified TX)

- TX Beam Characteristics
 - With phase and amplitude taper:
(Taylor 35dB taper in aperture => no power loss)
 - Directivity sum beam TX
 - TX sum Beam (one way, 3dB): ~23.5 dBi
 - Az and El: ~ +/- 6.5 deg



Lannik Psi *IFX radar*



● Concept (proposal)

TX beam

- Phase and amplitude taper

RX beam

- Uniform Taper (narrow sum beam)

Detector

- 2D CA-CFAR
 - Based on 4,5,4 beams covering the transmit beam (TX sum and diff)
 - That is, create 13 beam-dependent 2D CA-CFAR maps.

Estimator

- Range: Detected Range bin (no interpolation needed)
- Doppler: Detected Doppler bin (no interpolation needed)
- DOA: Monopulse in each beam based on phase gradient computed along u and v (i.e., will (ideally) independent of beam direction)

Tracker

- EKF enough?
- Association (GNN too much)
- Internal Radar-IMU

